# 1 What is in this repository?

This repository is a PyTorch research codebase for **video watch-time prediction**. It contains an implementation of the paper:

> *"Multi-Granularity Distribution Modeling for Video Watch Time Prediction via Exponential-Gaussian Mixture Network"*

The repository also includes several baseline models and code to preprocess the public KuaiRec dataset.

## 1.1 High-level structure

- `model/`: Model implementations (proposed EGMN and baselines).
- `dataloader/`: Dataset loaders that read preprocessed files and provide PyTorch `DataLoaders`.
- `dataset/kuairec/`: KuaiRec preprocessing scripts.
- `run_*.py`: Entrypoints to train/evaluate each method.
- `utils.py`: Evaluation metrics (MAE, XAUC, KL divergence) and helper utilities.

## 1.2 Code examples

**Typical entrypoint invocation**

```
# Train + evaluate the proposed EGMN model on KuaiRec
python run_egmn.py --dataset_name kuairec --device cuda:0

# Baselines (examples)
python run_cread.py --dataset_name kuairec
python run_d2q.py --dataset_name kuairec
```

**What the dataloader yields** The KuaiRec dataloader returns a tuple `(features_dict, label)`, where `features_dict` contains tensors keyed by feature name (e.g. `duration`, categorical IDs, and sequence masks).

```
# Pseudocode reflecting dataloader/kuairec.py
for features, label in dataloaders["train"]:
    # features is a dict of tensors, label is the normalized play_time
    y_hat = model.predict(features)
```

**Metrics used in evaluation**

```
# utils.py (simplified)
def eval_mae(labels, scores):
    return np.mean(np.abs(labels - scores))

def eval_xauc(labels, preds):
    # ranks by prediction, then measures label ordering agreement
    ...

def eval_kl(samples_p, samples_q, bins=100, epsilon=1e-10):
```

```
    hist_p, bin_edges = np.histogram(samples_p, bins=bins, density=True)
    hist_q, _ = np.histogram(samples_q, bins=bin_edges, density=True)
    ...
    return np.sum(hist_p * np.log(hist_p / hist_q))
```

### 1.3 Evaluation outputs

The run scripts train on `train` and evaluate on `test`, printing (at least):

- **MAE**: mean absolute error on watch-time prediction.
- **XAUC**: a ranking-style metric computed from label–prediction ordering.
- **KL**: KL divergence between binned empirical distributions.